

Steps to create redux-demo:

1. Give the command in terminal => npx create-react-app redux-demo
redux-demo folder and src folder are created.
 2. Next type the command => cd redux-demo.
 3. Now install all the packages.
 - npm install redux react-redux
 - npm install @reduxjs/toolkit react-redux
 - npm install redux-thunk
 - npm install redux-observable rxjs
 - npm install redux-undo
 - npm install immutable
 4. Give these commands one after other required packages will get installed.
 5. Now create directory features in src folder
 6. Then add postsSlice.js file in that.
- Run the file using command npm start.

postsSlice.js

```
// features/postsSlice.js
import { createSlice, createAsyncThunk } from '@reduxjs/toolkit';
// Step 1: Create async thunk
export const fetchPosts = createAsyncThunk('posts/fetchPosts', async () => {
  const response = await fetch('https://jsonplaceholder.typicode.com/posts');
  return response.json();
});
// Step 2: Create slice
const postsSlice = createSlice({
  name: 'posts',
  initialState: {
    items: [],
    status: 'idle', // idle | loading | succeeded | failed
    error: null,
  },
  reducers: {},
  extraReducers: (builder) => {
    builder
      .addCase(fetchPosts.pending, (state) => {
        state.status = 'loading';
      })
      .addCase(fetchPosts.fulfilled, (state, action) => {
        state.status = 'succeeded';
        state.items = action.payload;
      })
      .addCase(fetchPosts.rejected, (state, action) => {
        state.status = 'failed';
        state.error = action.error.message;
      });
  },
});
export default postsSlice.reducer;
```

store.js

```
// store.js
import { configureStore } from '@reduxjs/toolkit';
import postsReducer from '../features/postsSlice';

export const store = configureStore({
  reducer: {
    posts: postsReducer,
  },
});
```

App.js

```
// App.js
import React, { useEffect } from 'react';
import { useSelector, useDispatch } from 'react-redux';
import { fetchPosts } from '../features/postsSlice';

function App() {
  const dispatch = useDispatch();
  const { items, status, error } = useSelector((state) => state.posts);

  useEffect(() => {
    if (status === 'idle') {
      dispatch(fetchPosts());
    }
  }, [status, dispatch]);

  return (
    <div>
      <h1>Redux Async Data Fetching Example</h1>
      {status === 'loading' && <p>Loading...</p>}
      {status === 'failed' && <p>Error: {error}</p>}
      {status === 'succeeded' &&
        items.slice(0, 5).map((post) => (
          <div key={post.id}>
            <h3>{post.title}</h3>
            <p>{post.body}</p>
          </div>
        ))}
    </div>
  );
}

export default App;
```

App.test.js

```
import { render, screen } from '@testing-library/react';
import App from './App';

test('renders learn react link', () => {
  render(<App />);
  const linkElement = screen.getByText(/learn react/i);
  expect(linkElement).toBeInTheDocument();
});
```

index.js

```
// index.js
import React from 'react';
import ReactDOM from 'react-dom/client';
import { Provider } from 'react-redux';
import { store } from './store';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <Provider store={store}>
    <App />
  </Provider>
);
```